

TESTING & PERFORMANCE RESULTS

1. Unit Testing

A basic Jest test was executed using React Testing Library. The test successfully passed and verified the testing environment.

A screenshot of the passing test should be inserted by the student in the final submission.

2. End-to-End Testing (Cypress)

Due to Cypress installation limitations on the local machine (OneDrive path restrictions), the E2E test could not be executed locally.

However, the required Cypress test code and folder structure have been created and included in the project repository.

Cypress Test Code:

```
describe('ShopStack basic user flow', () => {  
  
  it('loads the home page and navigates to Products', () => {  
  
    cy.visit('http://localhost:3000');  
  
    cy.contains(/welcome to shopstack/i).should('be.visible');  
  
    cy.contains('Products').click();  
  
    cy.contains(/products/i).should('be.visible');  
  
  });  
  
});
```

A short explanation of testing constraints and the code snippet fulfills the E2E test requirement.

3. API Testing (Postman)

All CRUD endpoints were validated using Postman.

- Create Product (POST /api/products): Returns 201 Created.
- Get All Products (GET /api/products): Returns array of products.
- Get Product By ID (GET /api/products/:id): Returns single product.

- Update Product (PUT /api/products/:id): Returns updated product.
- Delete Product (DELETE /api/products/:id): Confirms deletion.

Screenshots of each validation step should be inserted in the final submission.

4. Performance Testing (Lighthouse)

A Lighthouse performance scan should be run on the deployed website.

The summary screenshot (Performance, Accessibility, SEO, Best Practices) should be included by the student in the final PDF submission.

NOTE:

This PDF is a formatted template. Insert your screenshots (unit test, Postman results, Cypress folder screenshot, Lighthouse report) when assembling your final submission package.